

Chapter 2

The Real and Complex Number Systems

In everyday life, knowingly or unknowingly, we are doing with numbers. Therefore, it will be nice if we get familiarized with numbers. Whatever course (which needs the concept of mathematics) we take, we face with the concept of numbers directly or indirectly. For this purpose, numbers and their basic properties will be introduced under this chapter.

Objective of the Chapter

At the end of this chapter, students will be able to:

- check the closure property of a given set of numbers on some operations
- determine the GCF and LCM of natural numbers
- apply the principle of mathematical induction to prove different mathematical formulae
- determine whether a given real number is rational number or not
- plot complex numbers on the complex plane
- convert a complex number from rectangular form to polar form and vice-versa
- extract roots of complex numbers

2.1 The real number System

2.1.1 The set of natural numbers

The history of numbers indicated that the first set of numbers used by the ancient human beings for counting purpose was the set of natural (counting) numbers.

Definition 2.1.1

The set of natural numbers is denoted by $\mathbf{N}$ and is described as $\mathbf{N} = \{ 1, 2, 3, \dots \}$

2.1.1.1 Operations on the set of natural numbers

i) Addition (+)

If two natural numbers a & b are added using the operation “+”, then the sum $a+b$ is also a natural number. If the sum of the two natural numbers a & b is denoted by c , then we can write the operation as: $c = a+b$, where c is called the sum and a & b are called terms.

Example: $3+8 = 11$, here 11 is the sum whereas 3 & 8 are terms.

ii) Multiplication ($\times$)

If two natural numbers a & b are multiplied using the operation “ $\times$ ”, then the product $a \times b$ is also a natural number. If the product of the two natural numbers a & b is denoted by c , then we can write the operation as: $c = a \times b$, where c is called the product and a & b are called factors.

Example 2.1.3: $3 \times 4 = 12$, here 12 is the product whereas 3 & 4 are factors.

Properties of addition and multiplication on the set of natural numbers

- i. For any two natural numbers a & b , the sum $a+b$ is also a natural number. For instance in the above example, 3 and 8 are natural numbers, their sum 11 is also a natural number. In general, we say that the set of natural numbers is closed under addition.
- ii. For any two natural numbers a & b , $a + b = b + a$.

Example 2.1.1: $3+8 = 8+3 = 11$. In general, we say that addition is commutative on the set of natural numbers.

- iii. For any three natural numbers a , b & c , $(a+b)+c = a+(b+c)$.

Example 2.1.2: $(3+8)+6 = 3+(8+6) = 17$. In general, we say that addition is associative on the set of natural numbers.

- iv. For any two natural numbers a & b , the product $a \times b$ is also a natural number. For instance in the above example, 3 and 4 are natural numbers, their product 12 is also a natural number. In general, we say that the set of natural numbers is closed under multiplication.
- v. For any two natural numbers a & b , $a \times b = b \times a$.

Example 2.1.4: $3 \times 4 = 4 \times 3 = 12$. In general, we say that multiplication is commutative on the set of natural numbers.

- vi. For any three natural numbers a , b & c , $(a \times b) \times c = a \times (b \times c)$.

Example 2.1.5: $(2 \times 4) \times 5 = 2 \times (4 \times 5) = 40$. In general, we say that multiplication is associative on the set of natural numbers.

- vi. For any natural number a , it holds that $a \times 1 = 1 \times a = a$.

Example 2.1.6: $6 \times 1 = 1 \times 6 = 6$. In general, we say that multiplication has an identity element on the set of natural numbers and 1 is the identity element.

- vii. For any three natural numbers a , b & c , $a \times (b+c) = (a \times b) + (a \times c)$.

Example 2.1.7: $3 \times (5+7) = (3 \times 5) + (3 \times 7) = 36$. In general, we say that multiplication is distributive over addition on the set of natural numbers.

Note: Consider two numbers a and b , we say a is greater than b denoted by $a > b$ if $a - b$ is positive.

2.1.1.2 Order Relation in N

i) Transitive property:

For any three natural numbers a , b & c , $a > b$ & $b > c \Rightarrow a > c$

ii) Addition property:

For any three natural numbers a , b & c , $a > b \Rightarrow a + c > b + c$

iii) Multiplication property:

For any three natural numbers a , b and c , $a > b \Rightarrow ac > bc$

iv) Law of trichotomy

For any two natural numbers a & b we have $a > b$ or $a < b$ or $a = b$.

2.1.1.3 Factors of a number

Definition 2.2

If $a, b, c \in N$ such that $ab = c$, then a & b are factors (divisors) of c and c is called product (multiple) of a & b .

Example 2.8: Find the factors of 15.

Solution: Factors of 15 are 1, 3, 5, 15. Or we can write it as: $F_{15} = \{1, 3, 5, 15\}$

Definition 2.3 A number $a \in N$ is said to be

- i. **Even** if it is divisible by 2.
- ii. **Odd** if it is not divisible by 2.
- iii. **Prime** if it has only two factors (1 and itself).
- iv. **Composite**: if it has three or more factors.

Example 2.9: 2, 4, 6, ... are even numbers

Example 2.10: 1, 3, 5, ... are odd numbers

Example 2.11: 2, 3, 5, ... are prime numbers

Example 2.12: 4, 6, 8, 9, ... are composite numbers

Remark: 1 is neither prime nor composite.

2.1.1.5 Prime Factorization

Definition 2.4

Prime factorization of a composite number is the product of all its prime factors.

Example 2.9:

a) $6 = 2 \times 3$ b) $30 = 2 \times 3 \times 5$ c) $12 = 2 \times 2 \times 3 = 2^2 \times 3$ d) $8 = 2 \times 2 \times 2 = 2^3$ e) $180 = 2^2 \times 3^2 \times 5$

Fundamental Theorem of Arithmetic:

Every composite number can be expressed as a product of its prime factors. This factorization is unique except the order of the factors.

2.1.1.6 Greatest Common Factor (GCF)

Definition 2.5

The greatest common factor (GCF) of two numbers a & b is denoted by $\text{GCF}(a, b)$ and is the greatest number which is a factor of each of the given number.

Note: If the GCF of two numbers is 1, then the numbers are called relatively prime.

Example 2.10: Consider the two numbers 24 and 60.

Now $F_{24} = \{ 1, 2, 3, 4, 6, 8, 12, 24 \}$

and $F_{60} = \{ 1, 2, 3, 4, 5, 6, 10, 12, 15, 20, 30, 60 \}$

Next $F_{24} \cap F_{60} = \{ 1, 2, 3, 4, 6, 12 \}$ from which 12 is the greatest.

Therefore, $\text{GCF}(24, 60) = 12$.

This method of finding the GCF of two or more numbers is usually lengthy and time consuming.

Hence an alternative method (Prime factorization method) is provided as below:

Step 1: Find the prime factorization of each of the natural numbers

Step 2: Form the GCF of the given numbers as the product of every factor that appears in each of the prime factorization but take the least number of times it appears.

Example 2.11: Consider the two numbers 24 and 60.

Step1: $24 = 2^3 \times 3$

$$60 = 2^2 \times 3 \times 5$$

Step 2: The factors that appear in both cases are 2 and 3, but take the numbers with the least number of times.

$$\therefore \text{GCF}(24, 60) = 2^2 \times 3 = \underline{\underline{12}}$$

Example 2.12: Consider the three numbers 20, 80 and 450.

Step1: $20 = 2^2 \times 5$

$$80 = 2^4 \times 5$$

$$450 = 2 \times 3^2 \times 5^2$$

Step 2: The factors that appear in all cases are 2 and 5, but take the numbers with the least number of times.

$$\therefore \text{GCF}(20, 80, 450) = 2 \times 5 = \underline{\underline{10}}$$

2.1.1.7 Least Common Multiple (LCM)

Definition 2.6

The least common multiple (LCM) of two numbers a & b is denoted by $\text{LCM}(a, b)$ and is the least number which is a multiple of each of the given number.

Example 2.13: Consider the two numbers 18 and 24.

Now $M_{18} = \{ 18, 36, 54, 72, 90, 108, 126, 144, \dots \}$

and $M_{24} = \{ 24, 48, 72, 96, 120, 144, \dots \}$

Next $M_{18} \cap M_{24} = \{ 72, 144, \dots \}$ from which 72 is the least.

Therefore, $\text{LCM}(18, 24) = 72$.

This method of finding the LCM of two or more numbers is usually lengthy and time consuming.

Hence an alternative method (Prime factorization method) is provided as below:

Step 1: Find the prime factorization of each of the natural numbers

Step 2: Form the LCM of the given numbers as the product of every factor that appears in any of the prime factorization but take the highest number of times it appears.

Example 2.14: Consider the two numbers 18 and 24.

Step1: $18 = 2^2 \times 3^2$

$$24 = 2^3 \times 3$$

Step 2: The factors that appear in any case are 2 and 3, but take the numbers with the highest number of times.

$$\therefore \text{LCM}(18, 24) = 2^3 \times 3^2 = \underline{\underline{72}}$$

Example 2.15: Consider the three numbers 20, 80 and 450.

Step1: $20 = 2^2 \times 5$

$$80 = 2^4 \times 5$$

$$450 = 2 \times 3^2 \times 5^2$$

Step 2: The factors that appear in any cases are 2, 3 and 5, but take the numbers with the highest number of times.

$$\therefore \text{LCM}(20, 80, 450) = 2^4 \times 3^2 \times 5^2 = \underline{\underline{3600}}$$

2.1.1.8 Well ordering Principle in the set of natural numbers

Proposition 2.7

Every non-empty subset of the set of natural numbers has smallest (least) element.

Example 2.16 $A = \{ 2, 3, 4, \dots \} \subseteq N$. smallest element of $A = 2$.

Note: The set of counting numbers including zero is called the set of whole numbers and is denoted by **W**. i.e $\mathbf{W} = \{ 0, 1, 2, 3, \dots \}$

2.1.1.9 Principle of Mathematical Induction

Mathematical induction is one of the most important techniques used to prove in mathematics. It is used to check conjectures about the outcome of processes that occur repeatedly according to definite patterns. We will introduce the technique with examples.

For a given assertion involving a natural number n , if

- i. the assertion is true for $n = 1$ (usually).
- ii. it is true for $n = k+1$, whenever it is true for $n = k$ ($k \geq 1$), then the assertion is true for every natural number n .

The method is used to prove different propositions involving positive integers using three steps:

Step1: Prove that T_k (usually T_1) holds true.

Step 2: Assume that T_k for $k = n$ is true.

Step 3: Show that T_k is true for $k = n+1$.

Example 2.17 Show that $1+3+5+\dots+(2n-1) = n^2$.

Proof:

Step1. For $n=1$, $1=1^2$ which is true.

Step2. Assume that it is true for $n=k$

$$\text{i.e. } 1+3+5+\dots+(2k-1) = k^2.$$

Step3. We should show that it is true for $n=k+1$.

$$\text{Claim: } 1+3+5+\dots+(2k-1)+(2k+1) = (k+1)^2$$

$$\text{Now } \underline{1+3+5+\dots+(2k-1)} + (2k+1) = k^2 + (2k+1)$$

$$= k^2 + 2k + 1$$

$$= \underline{(k+1)^2} \text{ which is the required result.}$$

$\therefore$ It is true for any natural number n .

Example 2.18 Show that $1+2+3+\dots+(n) = \frac{n(n+1)}{2}$.

Proof:

Step1. For $n=1$, $1 = \frac{1(1+1)}{2}$ which is true.

Step2. Assume that it is true for $n=k$

$$\text{i.e. } 1+2+3+\dots+(k) = \frac{k(k+1)}{2}.$$

Step3. We should show that it is true for $n=k+1$

$$\text{Claim: } 1 + 2 + 3 + \dots + (k) + (k + 1) = \frac{(k + 1)(k + 2)}{2}.$$

$$\begin{aligned} \text{Now } 1 + 2 + 3 + \dots + (k) + (k + 1) &= \frac{k(k + 1)}{2} + (k + 1) \\ &= \frac{k(k + 1) + 2(k + 1)}{2} \\ &= \frac{(k + 1)(k + 2)}{2} \text{ which is the required result.} \end{aligned}$$

$\therefore$ It is true for any natural number n .

Example 2.19 Show that $5^n + 6^n < 9^n$ for $n \geq 2$.

Proof:

Step1. For $n = 2$, $61 < 81$ which is true

Step2. Assume that it is true for $n = k$.

$$\text{i.e. } 5^k + 6^k < 9^k.$$

Step3. We should show that it is true for $n = k + 1$

$$\text{Claim: } 5^{k+1} + 6^{k+1} < 9^{k+1}.$$

$$\begin{aligned} \text{Now } 5^{k+1} + 6^{k+1} &= 5 \cdot 5^k + 6 \cdot 6^k < 6 \cdot 5^k + 6 \cdot 6^k \\ &= 6(5^k + 6^k) \\ &< 9(5^k + 6^k) \\ &< 9(9^k) = 9^{k+1} \\ \Rightarrow 5^{k+1} + 6^{k+1} &< 9^{k+1} \text{ which is the required format.} \end{aligned}$$

$\therefore$ It is true for any natural number $n \geq 2$.

2.1.2 The set of Integers

As the knowledge and interest of human beings increased, it was important and obligatory to extend the natural number system. For instance to solve the equation $x+1=0$, the set of natural numbers was not sufficient. Hence the set of integers was developed to satisfy such extended demands.

Definition 2.8

The set of integers is denoted by $\mathbf{Z}$ and described as $\mathbf{Z} = \{ \dots, -2, -1, 0, 1, 2, \dots \}$

2.1.2.1 Operations on the set of integers

i) Addition (+)

If two integers a & b are added using the operation “+”, then the sum $a+b$ is also an integer. If the sum of the two integers a & b is denoted by c , then we can write the operation as: $c = a+b$, where c is called the sum and a & b are called terms.

Example 2.20: $4+9 = 13$, here 13 is the sum whereas 4 & 9 are terms.

ii) Subtraction (–)

For any two integers a & b , the operation of subtracting b from a , denoted by $a-b$ is defined by $a-b = a+(-b)$. This means that subtracting b from a is equivalent to adding the additive inverse of b to a .

Example 2.21: $7-5 = 7+(-5) = 2$

iii) Multiplication (×)

If two integers a & b are multiplied using the operation “×”, then the product $a \times b$ is also an integer. If the product of the two integers a & b is denoted by c , then we can write the operation as: $c = a \times b$, where c is called the product and a & b are called factors.

Example 2.22: $4 \times 7 = 28$, here 28 is the product whereas 4 & 7 are factors.

Properties of addition and multiplication on the set of integers

- i. For any two integers a & b , the sum $a+b$ is also an integer. For instance in the above example, 4 and 9 are integers, their sum 13 is also an integer. In general, we say that the set of integers is closed under addition.
- ii. For any two integers a & b , $a+b = b+a$.

Example 2.23: $4+9 = 9+4 = 13$. In general, we say that addition is commutative on the set of integers.

- iii. For any three integers a , b & c , $(a+b)+c = a+(b+c)$.

Example 2.24: $(5+9)+8 = 5+(9+8) = 22$. In general, we say that addition is associative on the set of integers.

- iv. For any integer a , it holds that $a+0 = 0+a = a$.

Example 2.25: $7+0 = 0+7 = 7$. In general, we say that addition has an identity element on the set of integers and 0 is the identity element.

- v. For any integer a , it holds that $a+(-a) = -a+a = 0$.

Example 2.26: $4+-4 = -4+4 = 0$. In general, we say that every integer a has an additive inverse denoted by $-a$.

- vi. For any two integers a & b , the product $a \times b$ is also an integer. For instance in the above example, 4 and 7 are integers, their product 28 is also an integer. In general, we say that the set of integers is closed under multiplication.

vii. For any two integers a & b , $a \times b = b \times a$.

Example 2.27: $4 \times 7 = 7 \times 4 = 28$. In general, we say that multiplication is commutative on the set of integers.

viii. For any three integers a , b & c , $(a \times b) \times c = a \times (b \times c)$.

Example 2.28: $(3 \times 5) \times 4 = 3 \times (5 \times 4) = 60$. In general, we say that multiplication is associative on the set of integers.

ix. For any integer a , it holds that $a \times 1 = 1 \times a = a$.

Example 2.29: $5 \times 1 = 1 \times 5 = 5$. In general, we say that multiplication has an identity element on the set of integers and 1 is the identity element.

x. For any three integers a , b & c , $a \times (b + c) = (a \times b) + (a \times c)$.

Example 2.30: $4 \times (5 + 6) = (4 \times 5) + (4 \times 6) = 44$. In general, we say that multiplication is distributive over addition on the set of integers.

2.1.2.2 Order Relation in $\mathbb{Z}$

i) **Transitive property:** For any three integers a , b & c , $a > b$ & $b > c \Rightarrow a > c$

ii) **Addition property:** For any three integers a , b & c , $a > b \Rightarrow a + c > b + c$

iii) **Multiplication property:** For any three integers a , b and c , where $c > 0$, $a > b \Rightarrow ac > bc$

iv) **Law of trichotomy:** For any two integers a & b we have $a > b$ or $a < b$ or $a = b$.

Exercise 2.1

1. Find an odd natural number x such that $\text{LCM}(x, 40) = 1400$.
2. There are between 50 and 60 number of eggs in a basket. When Loza counts by 3's, there are 2 eggs left over. When she counts by 5's, there are 4 left over. How many eggs are there in the basket?
3. The GCF of two numbers is 3 and their LCM is 180. If one of the numbers is 45, then find the second number.
4. Using Mathematical Induction, prove the following:
 - a) $6^n - 1$ is divisible by 5, for $n \geq 0$.
 - b) $2^n \leq (n+1)!$, for $n \geq 0$
 - c) $x^n + y^n$ is divisible by $x + y$ for odd natural number $n \geq 1$.
 - d) $2 + 4 + 6 + \dots + 2n = n(n+1)$
 - e) $1^2 + 2^2 + 3^2 + \dots + n^2 = \frac{n(n+1)(2n+1)}{6}$

$$f) 1^3 + 2^3 + 3^3 + \dots + n^3 = \frac{n^2 (n+1)^2}{4}$$

$$g) \frac{1}{1 \times 2} + \frac{1}{2 \times 3} + \frac{1}{3 \times 4} + \dots + \frac{1}{n(n+1)} = \frac{n}{n+1}$$

2.1.3 The set of rational numbers

As the knowledge and interest of human beings increased with time, it was again necessary to extend the set of integers. For instance to solve the equation $2x+1=0$, the set of integers was not sufficient. Hence the set of rational numbers was developed to satisfy such extended needs.

Definition 2.9

Any number that can be expressed in the form $\frac{a}{b}$, where a and b are integers and $b \neq 0$, is called a rational number. The set of rational numbers denoted by Q is described by

$$Q = \left\{ \frac{a}{b} : a \text{ and } b \text{ are integers and } b \neq 0 \right\}.$$

Notes:

- From the expression $\frac{a}{b}$, a is called numerator and b is called denominator.
- A rational number $\frac{a}{b}$ is said to be in lowest form if $\text{GCF}(a, b) = 1$.

2.1.3.1 Operations on the set of rational numbers

i) Addition (+)

If two rational numbers a/b and c/d are added using the operation “+”, then the sum defined as

$$\frac{a}{b} + \frac{c}{d} = \frac{ad + bc}{bd} \text{ is also a rational number.}$$

Example 2.31: $\frac{1}{2} + \frac{3}{5} = \frac{11}{10}$

ii) Subtraction (–)

For any two rational numbers a/b & c/d , the operation of subtracting c/d from a/b , denoted by $a/b - c/d$ is defined by $a/b - c/d = a/b + (-c/d)$.

Example 2.32: $\frac{1}{2} - \frac{3}{5} = \frac{-1}{10}$

iii) Multiplication ($\times$)

If two rational numbers a/b and c/d are multiplied using the operation “ $\times$ ”, then the product defined as $\frac{a}{b} \times \frac{c}{d} = \frac{ac}{bd}$ is also a rational number.

Example 2.33: $\frac{1}{2} \times \frac{3}{5} = \frac{3}{10}$

iv) Division ($\div$)

For any two rational numbers a/b & c/d , dividing a/b by c/d is defined by

$$\frac{a}{b} \div \frac{c}{d} = \frac{a}{b} \times \frac{d}{c}, \quad c \neq 0.$$

Example 2.34: $\frac{1}{2} \div \frac{3}{5} = \frac{1}{2} \times \frac{5}{3} = \frac{5}{6}$

Properties of addition and multiplication on the set of rational numbers

Let a/b , c/d and e/f be three rational numbers, then

- i. The set of rational numbers is closed under addition and multiplication.
- ii. Addition and multiplication are both commutative on the set of rational numbers.
- iii. Addition and multiplication are both associative on the set of rational numbers.
- iv. **0** is the additive identity
i.e., $a/b + 0 = 0 + a/b = a/b$.
- v. Every rational number has an additive inverse.
i.e., $a/b + (-a/b) = -a/b + a/b = 0$.
- vi. **1** is the multiplicative identity
i.e., $a/b \times 1 = 1 \times a/b = a/b$.
- vii. Every non-zero rational number has a multiplicative inverse.
i.e., $a/b \times b/a = b/a \times a/b = 1$.

2.1.3.2 Order Relation in Q

i) Transitive property

For any three rational numbers a/b , c/d & e/f $a/b > c/d$ & $c/d > e/f \Rightarrow a/b > e/f$.

ii) Addition property

For any three rational numbers a/b , c/d & e/f $a/b > c/d \Rightarrow a/b + e/f > c/d + e/f$.

iii) Multiplication property

For any three rational numbers a/b , c/d , e/f and $e/f > 0$

$$a/b > c/d \Rightarrow (a/b)(e/f) > (c/d)(e/f).$$

iv) Law of trichotomy

For any two rational numbers a/b & c/d we have $a/b > c/d$ or $a/b < c/d$ or $a/b = c/d$.

2.1.3.3 Decimal representation of rational numbers

A rational number $\frac{a}{b}$ can be written in decimal form using long division.

2.1.3.3.1 Terminating decimals

Example 2.35: Express the fraction number $\frac{25}{4}$ in decimal form.

$$\text{Solution : } \frac{25}{4} = 6.25$$

2.1.3.3.2 Non-terminating periodic decimals

Example 2.36: Express the fraction number $\frac{25}{3}$ in decimal form.

$$\text{Solution : } \frac{25}{3} = 8.333\ldots$$

Now we will see how to convert decimal numbers in to their fraction forms. In earlier mathematics topics, we have seen that multiplying a decimal by 10 pushes the decimal point to the right by one position and in general, multiplying a decimal by 10^n pushes the decimal point to the right by n positions. We will use this fact for the succeeding topics.

2.1.3.4 Fraction form of decimal numbers

A rational number which is written in decimal form can be converted to a fraction form as $\frac{a}{b}$ in lowest (simplified) form, where a and b are relatively prime.

2.1.3.4.1 Terminating decimals

Consider any terminating decimal number d . Suppose d terminates n digits after the decimal point. d can be converted to its fraction form as below:

$$d = d \times 1 = d \times \frac{1}{1} = d \times \left(\frac{10^n}{10^n} \right).$$

Example 2.37: Convert the terminating decimal 3.47 to fraction form.

$$\text{Solution: } 3.47 = 3.47 \times \frac{10^2}{10^2} = \frac{347}{100}.$$

2.1.3.4.2 Non-terminating periodic decimals

Consider any non-terminating periodic decimal number d . Suppose d has k non-terminating digits and p terminating digits after the decimal point. d can be converted to its fraction form as below:

$$d = d \times 1 = d \times \frac{1}{1} = d \times \left(\frac{10^{k+p} - 10^k}{10^{k+p} - 10^k} \right).$$

Example 2.38: Convert the non-terminating periodic decimal $42.\overline{538}$ to fraction form.

Solution: $k = 1$, $p = 2$.

$$\therefore d = d \times 1 = d \times \frac{1}{1} = d \times \left(\frac{10^{k+p} - 10^k}{10^{k+p} - 10^k} \right) = 42.\overline{538} \times \left(\frac{10^3 - 10}{10^3 - 10} \right) = \frac{42538.\overline{38} - 425.\overline{38}}{1000 - 10} = \frac{42113}{990}.$$

Note: From the above two cases, we can conclude that both terminating decimals and non-terminating periodic decimals are rational numbers. (Why? Justify).

2.1.3.5 Non-terminating and non-periodic decimals

Some decimal numbers are neither terminating nor non-terminating periodic. Such types of numbers are called irrational numbers.

Example 2.39: $62.757757775\dots$

Example 2.40: Show that $\sqrt{2}$ is an irrational number.

Proof:

Suppose $\sqrt{2}$ is a rational number

$$\Rightarrow \sqrt{2} = \frac{a}{b}, \text{ where } GCF(a, b) = 1$$

$$\Rightarrow 2 = \frac{a^2}{b^2}$$

$$\Rightarrow a^2 = 2b^2 \dots\dots\dots(*)$$

$$\Rightarrow a^2 \text{ is even}$$

$$\Rightarrow a \text{ is even}$$

$$\Rightarrow a = 2n \dots\dots\dots(**)$$

Putting this in (*) we get :

$$\Rightarrow 4n^2 = 2b^2$$

$$\Rightarrow b^2 = 2n^2$$

$$\Rightarrow b^2 \text{ is even}$$

$$\Rightarrow b \text{ is even}$$

$$\Rightarrow b = 2m \dots\dots\dots(***)$$

From (**) and (***) we get a contradiction that $\text{GCF}(a, b) = 1$ which implies that $\sqrt{2}$ is not a rational number.

Therefore, $\sqrt{2}$ is an irrational number.

2.1.4 The set of real numbers

Definition 2.10

A number is called a real number if and only if it is either a rational number or an irrational number.

The set of real numbers denoted by $\mathbb{R}$ can be described as the union of the set of rational and irrational numbers. i.e $\mathbb{R} = \{x : x \text{ is a rational number or an irrational number}\}$.

There is a 1-1 correspondence between the set of real numbers and the number line (For each point in the number line, there is a corresponding real number and vice-versa).

2.1.4.1 Operations on the set of real numbers

i) Addition (+)

If two real numbers are added using the operation “+”, then the sum is also a real number.

ii) Subtraction (−)

For any two real numbers a & b , the operation of subtracting b from a , denoted by $a - b$ is defined by $a - b = a + (-b)$.

iii) Multiplication (×)

If two real numbers a and b are multiplied using the operation “×”, then the product defined as $a \times b = ab$ is also a real number.

iv) Division (÷)

For any two real numbers a & b , dividing a by b is defined by $a \div b = a \times \frac{1}{b}$, $b \neq 0$.

Properties of addition and multiplication on the set of real numbers

Let a , b and c be three real numbers, then

- i. The set of real numbers is closed under addition and multiplication.
- ii. Addition and multiplication are commutative on the set of real numbers.
- iii. Addition and multiplication are associative on the set of real numbers.
- iv. **0** is the additive identity
i.e., $a + \mathbf{0} = \mathbf{0} + a = a$.
- v. Every real number has an additive inverse.

i.e., $a + (-a) = -a + a = 0$.

vi. **1** is the multiplicative identity

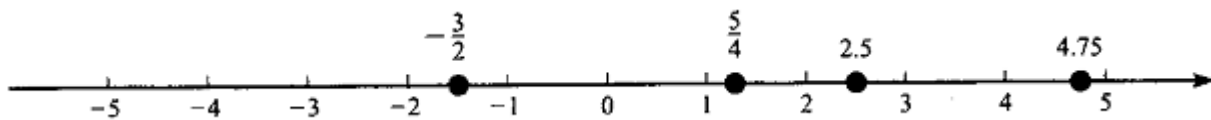
i.e., $a \times 1 = 1 \times a = a$.

vii. Every non-zero real number has a multiplicative inverse.

i.e., $a \times 1/a = 1/a \times a = 1$.

2.1.4.2 The real number and the number line

One of the most important properties of the real number is that it can be represented graphically by points on a straight line. The point 0 is termed as the origin. Points to the right of 0 are called positive real numbers and points to the left of 0 are called negative real numbers. Each point on the number line corresponds a unique real number and vice-versa.



Geometrically we say a is greater than b if a is located to the right of b on the number line.

2.1.4.3 Order Relation in R

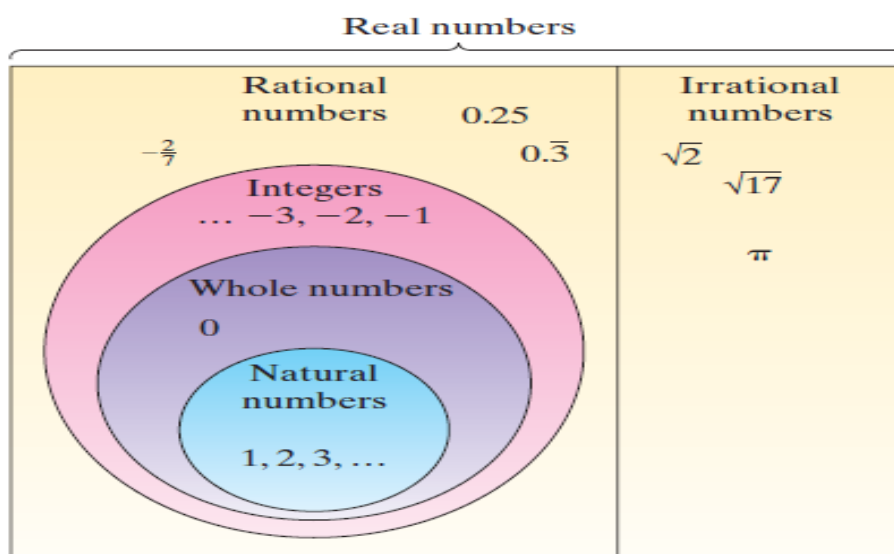
i) **Transitive property:** For any three real numbers a , b & c , $a > b$ & $b > c \Rightarrow a > c$.

ii) **Addition property:** For any three real numbers a , b & c , $a > b \Rightarrow a + c > b + c$.

iii) **Multiplication property:** For any three real numbers a , b , c and $c > 0$, we have $a > b \Rightarrow ac > bc$.

iv) **Law of trichotomy:** For any two real numbers a & b we have $a > b$ or $a < b$ or $a = b$.

Summary of the real number system



2.1.4.4 Intervals

Let a and b be two real numbers such that $a < b$, then the intervals which are subsets of $\mathbf{R}$ with end points a and b are denoted and defined as below:

- i. $(a, b) = \{x : a < x < b\}$ open interval from a to b .
- ii. $[a, b] = \{x : a \leq x \leq b\}$ closed interval from a to b .
- iii. $(a, b] = \{x : a < x \leq b\}$ open-closed interval from a to b .
- iv. $[a, b) = \{x : a \leq x < b\}$ closed-open interval from a to b .

2.1.4.5 Upper bounds and lower bounds

Definition 2.11

Let A be non – empty and $A \subseteq \mathbf{R}$.

- i. A point $a \in \mathbf{R}$ is said to be an upper bound of A iff $x \leq a$ for all $x \in A$.
- ii. An upper bound of A is said to be least upper bound (lub) iff it is the least of all upper bounds.
- iii. A point $a \in \mathbf{R}$ is said to be lower bound of A iff $x \geq a$ for all $x \in A$.
- ii. A lower bound of A is said to be greatest lower bound (glb) iff it is the greatest of all lower bounds.

Example 2.41 Consider the set $A = [2, 5) \subseteq \mathbf{R}$.

i) lower bounds are $\dots, -9, -3, 0, \frac{1}{2}, 1, 2$

Here the greatest element is 2.

$\therefore \text{glb} = 2$

ii) upper bounds are $5, 6, \frac{25}{3}, 20, 99, 1000 \dots$

Here the least element is 5.

$\therefore \text{lub} = 5$.

Example 2.42: Consider the set $A = \left\{ \frac{1}{n} \right\}$ for $n \in \mathbf{N}$.

Solution: $A = \left\{ 1, \frac{1}{2}, \frac{1}{3}, \dots \right\}$

i) lower bounds are $\dots, -3, -2, 0$

Here the greatest element is 0. Thus, $\text{glb} = 0$

ii) upper bounds are $1, 3, \frac{9}{2}, 50, \dots$

Here the least element is 1. Thus, $\text{lub} = 1$.

Based on the above definitions, we can define the completeness property of real numbers as below.

2.1.4.4 Completeness property of real number ($\mathbb{R}$)

Completeness property of real numbers states that: Every non-empty subset of $\mathbb{R}$ that has lower bounds has glb and every non-empty subset of $\mathbb{R}$ that has upper bounds has a lub.

Exercise 2.2

1. Express each of the following rational numbers as decimal:

a) $\frac{4}{9}$ b) $\frac{3}{25}$ c) $\frac{11}{7}$ d) $-5\frac{2}{3}$ e) $\frac{2}{77}$

2. Write each of the following as decimal and then as a fraction:

a) three tenths b) four thousands

3. Write each of the following in meters as a fraction and then as a decimal

a) 4mm b) 6cm and 4mm c) 56cm and 4mm

4. Classify each of the following as terminating or non-terminating periodic

a) $\frac{5}{13}$ b) $\frac{7}{10}$ c) $\frac{69}{64}$ d) $\frac{11}{60}$ e) $\frac{5}{12}$

5. Convert the following decimals to fractions:

a) $3.2\bar{5}$ b) $0.3\bar{14}$ c) $0.2\bar{75}$

6. Determine whether the following are rational or irrational:

a) $2.7\bar{5}$ b) $0.272727\ldots$ c) $\sqrt{8} - \frac{1}{\sqrt{2}}$

7. Which of the following statements are true and which of them are false?

- a) The sum of any two rational numbers is rational
- b) The sum of any two irrational numbers is irrational
- c) The product of any two rational numbers is rational
- d) The product of any two irrational numbers is irrational

11. Find two rational numbers between $\frac{1}{3}$ and $\frac{1}{2}$.

2.2 The set of complex numbers

The positive integers (natural numbers) were invented to count things. The negative integers were introduced to count money when we owed more than we had. The rational numbers were invented for measuring quantities. Since quantities like voltage, length and time can be measured using fractions, they can be measured using the rational numbers.

The real numbers were invented for wholly mathematical reasons: it was found that there were lengths such as the diagonal of the unit square which, in principle, couldn't be measured by the rational numbers, instead they can be measured using real numbers.

The complex numbers were invented for purely mathematical reasons, just like the real numbers and were intended to make things neat and tidy in solving equations. They were regarded with deep suspicion by the more conservative folk for a century. Complex numbers are points in the plane, together with a rule telling you how to multiply them. They are two-dimensional, whereas the real numbers are one dimensional.

Equations of the form $x^2 + 1 = 0$ has no solution on the set of real numbers. Therefore, the set of complex numbers permits us to solve such equations.

Definition 2.12

The set of complex numbers is denoted by $\mathbb{C}$ and is described by $\mathbb{C} = \{z / z = x + iy, x, y \in \mathbb{R} \text{ and } i^2 = -1\}$.

From the expression $z = x + iy$, x is called the real part and is denoted by $\text{Re}(Z)$

y is called the imaginary part and is denoted by $\text{Im}(Z)$.

Note: If $x = 0$, the number is called purely imaginary and if $y = 0$, the number is called purely real. Complex numbers can be defined as an order pair (x, y) of real numbers that can be interpreted as points in the complex plane (z - plane) with coordinates x and y .

Example 2.43: Find the real & imaginary part of the following complex numbers :

a) $z = 3 + 7i$

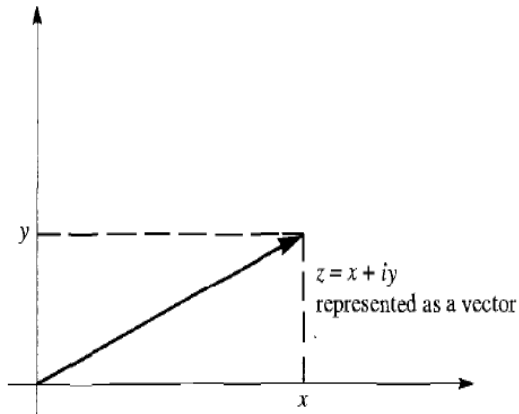
Solution: real part = 3 & imaginary part = 7

b) $z = 1 - i$

Solution: real part = 1 & imaginary part = -1

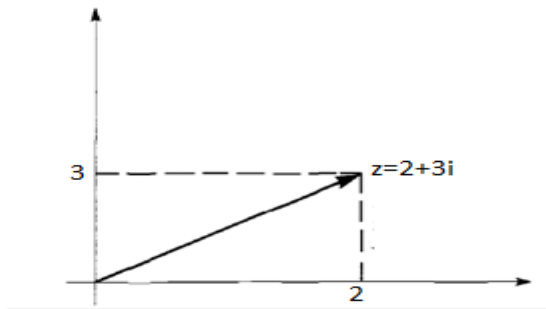
2.2.1 Plotting complex numbers

Any complex number $z = x + iy$ can be drawn in the complex plane as below :



Example 2.44: Draw the complex number $z = 2+3i$

Solution:



Equality of Complex numbers

Two complex numbers $z_1 = a + ib$ and $z_2 = c + id$ are equal iff $a = c$ & $b = d$.

Example 2.45 If $z_1 = 2 + ix$ and $z_2 = y + 6i$ are equal, then find the value of x & y .

Solution :

$$x = 6, y = 2.$$

2.2.2 Operations on Complex numbers

Let $z_1 = a + ib$ and $z_2 = c + id$ be any two complex numbers, then

i) $z_1 + z_2 = (a + c) + i(b + d)$

ii) $z_1 - z_2 = (a - c) + i(b - d)$

iii) $z_1 \cdot z_2 = (a + ib) \cdot (c + id) = a(c + id) + ib(c + id) = ac + iad + ibc - bd = (ac - bd) + i(ad + bc)$

iv) $\frac{z_1}{z_2} = \frac{(a + ib)}{(c + id)}, z_2 \neq 0.$

Example 2.46 If $z_1 = 2 + 3i$ and $z_2 = 4 + i$, then find a) $z_1 + z_2$ b) $z_1 - z_2$ c) $z_1 \cdot z_2$ d) $\frac{z_1}{z_2}$

Sol : a) $z_1 + z_2 = 6 + 4i$

b) $z_1 - z_2 = -2 + 2i$

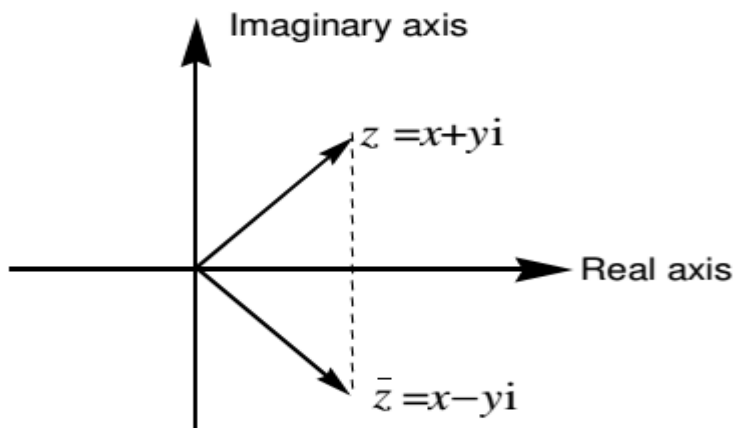
c) $z_1 \cdot z_2 = (2 + 3i) \cdot (4 + i) = 8 + 2i + 12i - 3 = 5 + 14i$

d) $\frac{z_1}{z_2} = \frac{2 + 3i}{4 + i}$

2.2.3 Conjugate of a complex number

Definition 2.13

The conjugate of a complex number $z = x + iy$ is denoted by $\bar{z}$ and is defined as $\bar{z} = x - iy$. It can be represented by the point $(x, -y)$ which is the reflection of the point (x, y) about the x-axis.



Example 2.47: Find the conjugate of the complex number $z = 2 + 9i$.

Solution :

$$z = 2 + 9i$$

$$\Rightarrow \bar{z} = 2 - 9i$$

Properties of Conjugate

$$a. \bar{\bar{z}} = z$$

$$b. z + \bar{z} = 2x = 2 \operatorname{Re}(z) = 2 \left(\frac{z + \bar{z}}{2} \right)$$

$$c. z - \bar{z} = 2iy = 2i \operatorname{Im}(z)$$

$$d. \overline{z_1 + z_2} = \bar{z}_1 + \bar{z}_2$$

$$e. \overline{z_1 - z_2} = \bar{z}_1 - \bar{z}_2$$

$$f. \overline{z_1 \cdot z_2} = \bar{z}_1 \cdot \bar{z}_2 \quad g. \overline{\left(\frac{z_1}{z_2} \right)} = \frac{\bar{z}_1}{\bar{z}_2}$$

proof : a) let $z = x + iy$

$$\Rightarrow \bar{z} = x - iy$$

$$\Rightarrow \overline{\bar{z}} = x + iy$$

$$\therefore \bar{\bar{z}} = z.$$

d) Let $z_1 = x_1 + iy_1$ & $z_2 = x_2 + iy_2$

$$\Rightarrow \bar{z}_1 = x_1 - iy_1 \quad \& \quad \bar{z}_2 = x_2 - iy_2$$

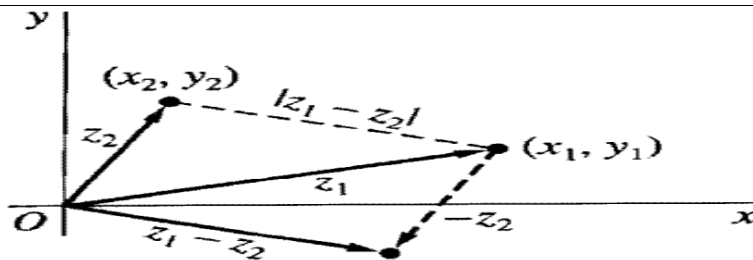
$$\begin{aligned} \text{Now } \overline{z_1 + z_2} &= \overline{(x_1 + iy_1) + (x_2 + iy_2)} = \overline{(x_1 + x_2) + i(y_1 + y_2)} = (x_1 + x_2) - i(y_1 + y_2) \\ &= x_1 - iy_1 + x_2 - iy_2 \\ &= \bar{z}_1 + \bar{z}_2 \end{aligned}$$

The others are left for the reader.

2.2.4 Modulus (Norm) of a complex number

Definition 2.14

The modulus of a complex number $z = x + iy$ is a non-negative real number denoted by $|z|$ and is defined as $|z| = \sqrt{x^2 + y^2}$. Geometrically, the number $|z|$ represents the distance between the point (x, y) and the origin.



Example 2.48: Find the modulus of the complex number $z = 3 - 4i$.

Solution : $z = 3 - 4i$

$$|z| = \sqrt{(3)^2 + (-4)^2} = \sqrt{25} = 5$$

Properties of modulus

a. $|z| = |\bar{z}|$

b. $|z|^2 = z \cdot \bar{z}$

c. $|z_1 \cdot z_2| = |z_1| \cdot |z_2|$

d. $\left| \frac{z_1}{z_2} \right| = \frac{|z_1|}{|z_2|}$

e. $|z_1 + z_2| \leq |z_1| + |z_2|$ triangle inequality

f. $|z_1 - z_2| \geq ||z_1| - |z_2||$

proof (a) let $z = x + iy$ from which $\bar{z} = x - iy$

$$\Rightarrow |z| = \sqrt{x^2 + y^2} \quad \Rightarrow |\bar{z}| = \sqrt{x^2 + y^2}$$

$$\therefore |z| = |\bar{z}|.$$

proof (b) let $z = x + iy$ from which $\bar{z} = x - iy$

$$\Rightarrow |z|^2 = x^2 + y^2$$

Now $z \cdot \bar{z} = (x + iy)(x - iy) = x^2 + y^2$

$$\therefore |z|^2 = z \cdot \bar{z}$$

proof (c) $|z_1 \cdot z_2|^2 = (z_1 \cdot z_2)(\overline{z_1 \cdot z_2}) = (z_1 \cdot z_2)(\bar{z}_1 \cdot \bar{z}_2) = z_1 \cdot \bar{z}_1 \cdot z_2 \cdot \bar{z}_2$

$$= |z_1|^2 \cdot |z_2|^2$$

$$= (|z_1| \cdot |z_2|)^2 \dots\dots\dots \text{from (b)}$$

$$\Rightarrow |z_1 \cdot z_2| = |z_1| \cdot |z_2|$$

proof (d) $\left| \frac{z_1}{z_2} \right|^2 = \frac{z_1(\bar{z}_1)}{z_2(\bar{z}_2)} = \frac{|z_1|^2}{|z_2|^2} = \left(\frac{|z_1|}{|z_2|} \right)^2$

$$\Rightarrow \left| \frac{z_1}{z_2} \right| = \frac{|z_1|}{|z_2|}.$$

The others are left for the reader.

2.2.5 Additive and multiplicative inverses

Let $z = x + iy$ be a complex number, then

i) its additive inverse denoted by $(-z)$ is given by: $-z = -(x + iy) = -x - iy$.

ii) its multiplicative inverse denoted by z^{-1} is given by: $z^{-1} = \frac{1}{x + iy} = \frac{x}{x^2 + y^2} - \frac{iy}{x^2 + y^2}$.

Example 2.49: Find the additive and the multiplicative inverse of $z = 3 + 4i$.

Solution: $z = 3 + 4i$

i) $-z = -3 - 4i$

ii) $z^{-1} = \frac{1}{3 + 4i} = \frac{1}{3 + 4i} \cdot \frac{3 - 4i}{3 - 4i} = \frac{3}{25} - \frac{4i}{25}$.

Exercise 2.3

1. Verify that

a) $(\sqrt{2} - i) - i(1 - \sqrt{2}i) = -2i$

b) $(2, -3)(-2, 1) = (-1, 8)$

c) $(3, 1)(3, -1)\left(\frac{1}{5}, \frac{1}{10}\right) = (2, 1)$

d) $(2 + 3i)^2 - (3i - 6) = 1 + 9i$

2. Show that

a) $\operatorname{Re}(iz) = -\operatorname{Im}(z)$

b) $\operatorname{Im}(iz) = \operatorname{Re}(z)$

c) $(z+1)^2 = z^2 + 2z + 1$

3. Do the following operations and simplify your answer.

a) $\frac{1+2i}{3-4i} + \frac{2-i}{5i}$

b) $\frac{5i}{(1-i)(2-i)(3-i)}$

c) $(1-i)^3$

4. Locate the complex numbers z_1+z_2 and z_1-z_2 , as vectors where

a) $z_1 = 2i, z_2 = \frac{2}{3} - i$

b) $z_1 = (-\sqrt{3}, 1), z_2 = (\sqrt{3}, 0)$

c) $z_1 = (-3, 1), z_2 = (1, 4)$

d) $z_1 = a + ib, z_2 = a - ib$

5. Sketch the following set of points determined by the condition given below:

a) $|z - 1 + i| = 1$

b) $|z + i| \leq 3$

c) $|z - 4i| \geq 4$

6. Using properties of conjugate and modulus, show that

a) $\overline{\overline{z} + 3i} = z - 3i$

b) $\overline{iz} = -i \overline{z}$

c) $\overline{(2+i)^2} = 3 - 4i$

7. Show that $(-1+i)^7 = 8(-1-i)$.

8. Using mathematical induction, show that (when $n = 2, 3, \dots$)

a) $\overline{z_1 + z_2 + \dots + z_n} = \overline{z_1} + \overline{z_2} + \dots + \overline{z_n}$

b) $\overline{z_1 z_2 \dots z_n} = \overline{z_1} \overline{z_2} \dots \overline{z_n}$

9. Show that the equation $|z - z_0| = r$ which is a circle of radius r centered at z_0 can be written as $|z|^2 - 2\operatorname{Re}(z\overline{z_0}) + |z_0|^2 = r^2$.

2.2.6 Argument (Amplitude) of a complex number

Definition 2.15

Argument of a complex number $z = x+iy$ is the angle formed by the complex number $z = x+iy$ with the positive x-axis. The argument of a complex number $z = x+iy$ is denoted by **argz** and is given by $\arg(z) = \tan^{-1}\left(\frac{y}{x}\right)$.

The particular argument of z that lies in the range $-\pi < \theta \leq \pi$ is called the principal argument of z and is denoted by **Argz**.

Notes : i) $\operatorname{Arg} z \in (-\pi, \pi]$

ii) If $0 \leq \operatorname{Arg} z \leq \pi$, move counter clock wise direction, if not move the other direction.

Example 2.50: Find the principal argument of the following complex numbers:

a) $z = 1 + i$

b) $z = -2 + 2\sqrt{3}i$

c) $z = -\sqrt{3} - i$

Sol : a) $z = 1 + i$

$$\text{Arg} z = \tan^{-1}\left(\frac{1}{1}\right) = \tan^{-1}(1) = \frac{\pi}{4}$$

b) $z = -2 + 2\sqrt{3}i$

$$\text{Arg} z = \tan^{-1}\left(\frac{2\sqrt{3}}{-2}\right) = \tan^{-1}(-\sqrt{3}) = \frac{2\pi}{3}$$

c) $z = -\sqrt{3} - i$

$$\text{Arg} z = \tan^{-1}\left(\frac{-1}{-\sqrt{3}}\right) = \tan^{-1}\left(\frac{1}{\sqrt{3}}\right) = \frac{-5\pi}{6}$$

Properties of Arguments

i) $\text{Arg}(z_1 \cdot z_2) = \text{Arg} z_1 + \text{Arg} z_2$ ii) $\text{Arg}\left(\frac{z_1}{z_2}\right) = \text{Arg} z_1 - \text{Arg} z_2$

Example 2.51: Find the principal argument of a) $(1+i)(-1-i)$ b) $\left(\frac{-2+2i}{1-i}\right)$

Solution:

a) $\text{Arg}(1+i)(-1-i) = \text{Arg}(1+i) + \text{Arg}(-1-i) = \frac{\pi}{4} + \left(-\frac{3\pi}{4}\right) = \underline{\underline{-\frac{\pi}{2}}}$

b) $\text{Arg}\left(\frac{-2+2i}{1-i}\right) = \text{Arg}(-2+2i) - \text{Arg}(1-i) = \frac{3\pi}{4} - \left(-\frac{\pi}{4}\right) = \frac{4\pi}{4} = \underline{\underline{\pi}}$

2.2.7 Polar form of a complex number

Definition 2.16

Let r and θ be polar coordinates of the point (x, y) of the complex number $z = x+iy$. Since $x = r \cos \theta$ and $y = r \sin \theta$, then the complex number can be written as : $z = r(\cos \theta + i \sin \theta)$ which is called polar form, where r is modulus of z and θ is principal argument of z .

Example 2.52: Express the following complex numbers in polar form:

a) $z = 1 + i$

solution : $r = \sqrt{2}$ and $\theta = \tan^{-1}(1) = \frac{\pi}{4}$. Thus, $z = \sqrt{2}(\cos \frac{\pi}{4} + i \sin \frac{\pi}{4})$.

b) $z = 3 - 3i$

solution : $r = \sqrt{18}$ and $\theta = \tan^{-1}(-1) = -\pi/4$.

Thus, $z = \sqrt{18}(\cos -\pi/4 + i \sin -\pi/4) = \sqrt{18}(\cos \pi/4 - i \sin \pi/4)$.

Multiplication and division in polar forms

If $z_1 = r_1(\cos \theta_1 + i \sin \theta_1)$ and $z_2 = r_2(\cos \theta_2 + i \sin \theta_2)$, then

a) $z_1 \cdot z_2 = r_1 \cdot r_2 (\cos(\theta_1 + \theta_2) + i \sin(\theta_1 + \theta_2))$ b) $\frac{z_1}{z_2} = \frac{r_1}{r_2} (\cos(\theta_1 - \theta_2) + i \sin(\theta_1 - \theta_2))$.

Proof:

a) $z_1 \cdot z_2 = r_1(\cos \theta_1 + i \sin \theta_1) \cdot r_2(\cos \theta_2 + i \sin \theta_2)$
 $= r_1 \cdot r_2 [\cos \theta_1 (\cos \theta_2 + i \sin \theta_2) + i \sin \theta_1 (\cos \theta_2 + i \sin \theta_2)]$
 $= r_1 \cdot r_2 [\cos \theta_1 \cos \theta_2 + i \cos \theta_1 \sin \theta_2 + i \sin \theta_1 \cos \theta_2 - \sin \theta_1 \sin \theta_2]$
 $= r_1 \cdot r_2 [\cos \theta_1 \cos \theta_2 - \sin \theta_1 \sin \theta_2 + i (\cos \theta_1 \sin \theta_2 + \sin \theta_1 \cos \theta_2)]$
 $= \underline{\underline{r_1 \cdot r_2 [\cos(\theta_1 + \theta_2) + i \sin(\theta_1 + \theta_2)]}}$

b) $\frac{z_1}{z_2} = \frac{r_1(\cos \theta_1 + i \sin \theta_1)}{r_2(\cos \theta_2 + i \sin \theta_2)} = \frac{r_1}{r_2} \left(\frac{\cos \theta_1 + i \sin \theta_1}{\cos \theta_2 + i \sin \theta_2} \right)$
 $= \frac{r_1}{r_2} \left(\frac{\cos \theta_1 + i \sin \theta_1}{\cos \theta_2 + i \sin \theta_2} \right) \cdot \frac{\cos \theta_2 - i \sin \theta_2}{\cos \theta_2 - i \sin \theta_2}$
 $= \frac{r_1}{r_2} \left(\frac{\cos \theta_1 \cos \theta_2 - i \cos \theta_1 \sin \theta_2 + i \sin \theta_1 \cos \theta_2 + \sin \theta_1 \sin \theta_2}{\cos^2 \theta_2 + \sin^2 \theta_2} \right)$
 $= \frac{r_1}{r_2} \left(\frac{\cos \theta_1 \cos \theta_2 + \sin \theta_1 \sin \theta_2 + i (\sin \theta_1 \cos \theta_2 - \cos \theta_1 \sin \theta_2)}{1} \right)$
 $= \underline{\underline{\frac{r_1}{r_2} [\cos(\theta_1 - \theta_2) + i \sin(\theta_1 - \theta_2)]}}$

Example 2.53: If $z_1 = 6(\cos \pi/2 + i \sin \pi/2)$ and $z_2 = 2(\cos \pi/3 + i \sin \pi/3)$, then find

a) $z_1 \cdot z_2$ b) $\frac{z_1}{z_2}$

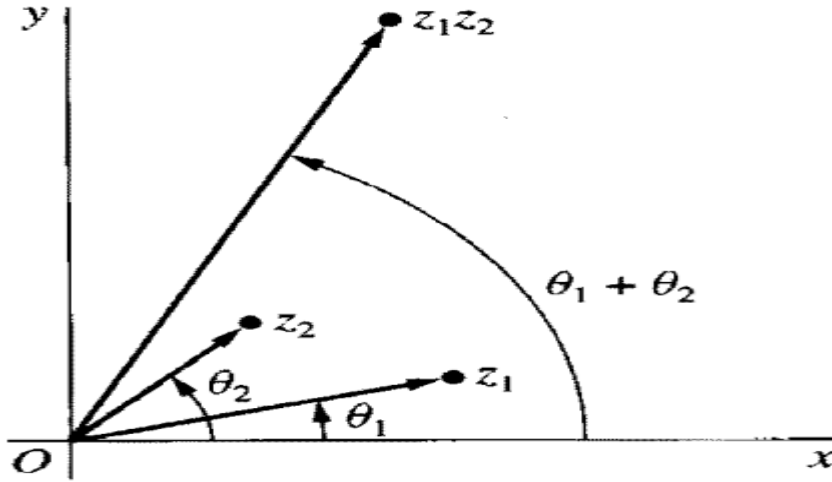
Solution :

a) $z_1 \cdot z_2 = 6 \cdot 2 [\cos(\pi/2 + \pi/3) + i \sin(\pi/2 + \pi/3)]$
 $= \underline{\underline{12 [\cos 5\pi/6 + i \sin 5\pi/6]}}$

b) $\frac{z_1}{z_2} = \frac{6}{2} [\cos(\pi/2 - \pi/3) + i \sin(\pi/2 - \pi/3)]$
 $= \underline{\underline{3 [\cos \pi/6 + i \sin \pi/6]}}$

- **Argument of a product**

The argument of the product of two complex numbers is the sum of their arguments.



Proof:

Let $z_1 = r_1(\cos \theta_1 + i \sin \theta_1)$ and $z_2 = r_2(\cos \theta_2 + i \sin \theta_2)$.

$$\begin{aligned}
 \text{Now } z_1 \cdot z_2 &= r_1(\cos \theta_1 + i \sin \theta_1) \cdot r_2(\cos \theta_2 + i \sin \theta_2) \\
 &= r_1 \cdot r_2 [\cos \theta_1 (\cos \theta_2 + i \sin \theta_2) + i \sin \theta_1 (\cos \theta_2 + i \sin \theta_2)] \\
 &= r_1 \cdot r_2 [\cos \theta_1 \cos \theta_2 + i \cos \theta_1 \sin \theta_2 + i \sin \theta_1 \cos \theta_2 - \sin \theta_1 \sin \theta_2] \\
 &= r_1 \cdot r_2 [\cos \theta_1 \cos \theta_2 - \sin \theta_1 \sin \theta_2 + i (\cos \theta_1 \sin \theta_2 + \sin \theta_1 \cos \theta_2)] \\
 &= \underline{\underline{r_1 \cdot r_2 [\cos(\theta_1 + \theta_2) + i \sin(\theta_1 + \theta_2)]}}
 \end{aligned}$$

$$\begin{aligned}
 \Rightarrow \arg(z_1 \cdot z_2) &= \tan^{-1} \left(\frac{\sin(\theta_1 + \theta_2)}{\cos(\theta_1 + \theta_2)} \right) \\
 &= \tan^{-1}(\tan(\theta_1 + \theta_2)) \\
 &= \underline{\underline{\theta_1 + \theta_2}}
 \end{aligned}$$

- **Argument of a quotient**

The argument of the quotient of two complex numbers is the difference of their arguments.

Proof:

$$\arg\left(\frac{z_1}{z_2}\right) = \arg(z_1 \cdot z_2^{-1}) = \arg(z_1) + \arg(z_2^{-1}) = \arg(z_1) + -(\arg z_2) = \arg(z_1) - \arg(z_2)$$

Example 2.54: $\arg\left(\frac{-4}{1+\sqrt{3}i}\right) = \arg(-4) - \arg(1+\sqrt{3}i) = \pi - \frac{\pi}{3} = \frac{2\pi}{3}$

De Moivre's Formula

recall the product: $z_1 \cdot z_2 = r_1 \cdot r_2 [\cos(\theta_1 + \theta_2) + i \sin(\theta_1 + \theta_2)]$.

Similarly, we get $z_1 \cdot z_2 \cdots z_n = r_1 \cdot r_2 \cdots r_n [\cos(\theta_1 + \theta_2 + \cdots + \theta_n) + i \sin(\theta_1 + \theta_2 + \cdots + \theta_n)]$.

Now we can generalize that $z^n = r^n [\cos(\theta + \theta + \cdots + \theta) + i \sin(\theta + \theta + \cdots + \theta)]$
 $= \underline{\underline{r^n [\cos n\theta + i \sin n\theta]}}$ which is called De Moivre's formula.

Example 2.55: Express $(2 + 2i)^{100}$ in polar form.

Solution: Let $z = 2 + 2i$. Then, $r = \sqrt{8}$, $\theta = \pi/4$ and hence

$$\begin{aligned} (2 + 2i)^{100} &= z^{100} \\ &= \sqrt{8}^{100} [\cos 100(\pi/4) + i \sin 100(\pi/4)] \\ &= \underline{\underline{8^{50} [\cos 25\pi + i \sin 25\pi]}}. \end{aligned}$$

Example 2.56: Express $(\sqrt{3} + i)^{60}$ in polar form.

Solution: Let $z = \sqrt{3} + i$. Then, $r = 2$, $\theta = \pi/6$.

$$\begin{aligned} \therefore (\sqrt{3} + i)^{60} &= z^{60} \\ &= 2^{60} [\cos 60(\pi/6) + i \sin 60(\pi/6)] \\ &= \underline{\underline{2^{60} [\cos 10\pi + i \sin 10\pi]}}. \end{aligned}$$

Euler's formula

The complex number $z = r(\cos \theta + i \sin \theta)$ can be written in exponential form as: $z = re^{i\theta}$ which is called Euler's formula.

Note: $z^n = r^n (\cos n\theta + i \sin n\theta) = r^n e^{i(n\theta)}$

Example 2.57: Express the complex number $z = 1+i$ using Euler's formula.

Solution: $z = 1+i$

$$\text{Now } r = \sqrt{2} \text{ \& } \theta = \pi/4 \Rightarrow z = re^{i\theta} = \underline{\underline{\sqrt{2}e^{i\pi/4}}}$$

Example 2.58: Express the complex number $z = 1 + \sqrt{3}i$ using Euler's formula.

Solution: $z = 1 + \sqrt{3}i$

Now $r = 2$ & $\theta = \pi/3 \Rightarrow z = re^{i\theta} = \underline{\underline{2e^{\pi i/3}}}$

Example 2.59: Express the complex number $z = (\sqrt{3} + i)^7$ using Euler's formula.

Solution: $z = (\sqrt{3} + i)^7$

Now $r = 2$, $\theta = \pi/6 \Rightarrow (\sqrt{3} + i)^7 = 2^7 e^{\frac{7\pi i}{6}} = 128e^{\frac{7\pi i}{6}}$

2.2.8 Extraction of roots

Suppose $z_o = r_o e^{i\theta_o}$ is the n^{th} root of a non-zero complex number $z = re^{i\theta}$, where $n \geq 2$.

Then, $z_o^n = z$, which implies that $r_o^n e^{in\theta_o} = re^{i\theta}$

$$\Rightarrow r_o^n = r \quad \& \quad n\theta_o = \theta + 2k\pi, \quad k = 0, 1, 2, \dots, (n-1).$$

$$\Rightarrow r_o = (r)^{1/n} \quad \& \quad \theta_o = \frac{\theta}{n} + \frac{2k\pi}{n}$$

$$\therefore z_o = \underline{\underline{(r)^{1/n} (e^{i(\frac{\theta}{n} + \frac{2k\pi}{n})})}} \text{ which is the } n^{\text{th}} \text{ root of } z, \text{ where } n = 2, 3, \dots \text{ and } k = 0, 1, 2, \dots, (n-1)$$

or we can denote it by C_k as : $C_k = \underline{\underline{(r)^{1/n} (e^{i(\frac{\theta}{n} + \frac{2k\pi}{n})})}}$, $k = 0, 1, 2, \dots, (n-1)$

Example 2.60: Find the square roots of the complex number $z = 1 + \sqrt{3}i$.

Solution:

$$z = 1 + \sqrt{3}i$$

Here $r = 2$, $\theta = \pi/3$

Hence $C_k = (r)^{1/n} (e^{i(\frac{\theta}{n} + \frac{2k\pi}{n})})$, $n = 2$, $k = 0, 1$.

$$\Rightarrow C_k = (2)^{1/2} (e^{i(\frac{\pi/3}{2} + \frac{2k\pi}{2})})$$

$$\Rightarrow C_k = \sqrt{2} (e^{i(\pi/6 + k\pi)})$$

$$\text{i.) If } k = 0, C_o = \sqrt{2} (e^{i(\pi/6)}) = \sqrt{2} (\cos \pi/6 + i \sin \pi/6) = \sqrt{2} (\frac{\sqrt{3}}{2} + \frac{i}{2}) = \frac{\sqrt{2}}{2} (\sqrt{3} + i) = \underline{\underline{\frac{\sqrt{6}}{2} + \frac{\sqrt{3}i}{2}}}$$

$$\text{ii) If } k = 1, C_1 = \sqrt{2} (e^{i(\pi/6 + \pi)}) = \sqrt{2} (\cos 7\pi/6 + i \sin 7\pi/6) = \sqrt{2} (-\frac{\sqrt{3}}{2} - \frac{i}{2}) = \frac{-\sqrt{2}}{2} (\sqrt{3} + i) = \underline{\underline{-\frac{\sqrt{6}}{2} - \frac{\sqrt{3}i}{2}}}$$

$$\therefore \text{ The square roots of } 1 + \sqrt{3}i \text{ are } C_o = \frac{\sqrt{6}}{2} + \frac{\sqrt{3}i}{2} \quad \& \quad C_1 = -\frac{\sqrt{6}}{2} - \frac{\sqrt{3}i}{2}.$$

Example 2.61: Find the cube roots of the complex number $z = 8i$.

Solution : We have $z = 8i$. Here $r = 8$, $\theta = \pi/2$, $n = 3$, $k = 0, 1, 2$.

$$\text{Hence, } C_k = (r)^{1/n} (e^{i(\frac{\theta}{n} + \frac{2k\pi}{n})})$$

$$\Rightarrow C_k = (8)^{1/3} (e^{i(\frac{\pi/2}{3} + \frac{2k\pi}{3})})$$

$$\Rightarrow C_k = 2(e^{i(\frac{\pi/6}{} + \frac{2k\pi}{3})})$$

$$\text{i) If } k = 0, C_0 = 2(e^{i(\pi/6)}) = (\cos \pi/6 + i \sin \pi/6) = 2(\frac{\sqrt{3}}{2} + \frac{i}{2}) = \underline{\underline{\sqrt{3} + i}}$$

$$\text{ii) If } k = 1, C_1 = 2(e^{i(\pi/6 + 2\pi/3)}) = 2(\cos 5\pi/6 + i \sin 5\pi/6) = 2(-\frac{\sqrt{3}}{2} + \frac{i}{2}) = \underline{\underline{-\sqrt{3} + i}}$$

$$\text{iii) If } k = 2, C_2 = 2(e^{i(\pi/6 + 4\pi/3)}) = 2(\cos 3\pi/2 + i \sin 3\pi/2) = 2(0 + -i) = \underline{\underline{-2i}}$$

$\therefore$ The cube roots of $8i$ are $C_0 = \sqrt{3} + i$, $C_1 = -\sqrt{3} + i$ & $C_2 = -2i$.

Exercise 2.4

1. Find the argument of the following complex numbers:

$$\text{a) } z = \frac{3i}{-1-i}$$

$$\text{b) } z = (\sqrt{3} - i)^6$$

2. Show that a) $|e^{i\theta}| = 1$

$$\text{b) } \overline{e^{i\theta}} = e^{-i\theta}$$

3. Using mathematical induction, show that $e^{i\theta_1} \cdot e^{i\theta_2} \cdot \dots \cdot e^{i\theta_n} = e^{i(\theta_1 + \theta_2 + \dots + \theta_n)}$, $n = 2, 3, \dots$

4. Show that a) $\cos 3\theta = \cos^3 \theta - 3\cos \theta \sin^2 \theta$

$$\text{b) } \sin 3\theta = 3\cos^2 \theta \sin \theta - \sin^3 \theta$$

5. Show that $1 + z + z^2 + \dots + z^n = \frac{1 - z^{n+1}}{1 - z}$, for $z \neq 1$.

6. Find the square roots of $z = 9i$

7. Find the cube roots of $z = -8i$

8. Solve the following equations:

$$\text{a) } z^{3/2} = 8i \quad \text{b) } z^2 + 4i = 0 \quad \text{c) } z^2 - 4i = 0$$